CS 577: Principles and Techniques of Data Science

Phase 3 (Group): Composite Visualization Report and Model Case

Fall 2026 | San Diego State University | 100 points

Overview

Your group notebook has seven parts:

Part	What you do	Points
0	Proposal revisit: what changed from Phase 2, how, and why	10
1	Composite portfolio: combine both members' plots and upgrade the two weakest	20
2	Lie detector: swap misleading plots, detect the trick, and write a checklist	20
3	Hero figure: the single best plot for your research question	10
4	Model case: regression or classification, linear or forest, and why	25
5	Honest data story	10
6	Reproducibility, README, and contributions	5
	Total	**100**

What Changed from Phase 2

- **Phase 3 is now the visualization phase.** Phase 3 was originally going to be modeling. Modeling has now started in lecture, so it moves to Phase 4. This phase turns your cleaned data into honest visual evidence and ends with a written case for the kind of model you will build next.
- **Phase 3 is split into two assignments.** Each member first completes the Phase 3 Individual assignment (20 points). This group assignment (100 points) is a composite of both individual notebooks: you combine them, improve them, and build on them. You do not redo the individual work.
- **The plot menu replaces the fixed chart list.** Phase 2 asked for a bar chart, a grouped bar chart, a histogram, and a trend or top N chart. Phase 3 uses every slot of the menu from lecture.
- **`iplot()` is no longer required.** Use seaborn and matplotlib (as in lecture) or `plotly.express`. `iplot()` is still allowed if your setup works.
- **Your dataset carries forward unless you have permission.** You continue from `phase2_working.csv`. Switching to a different dataset is allowed only with the instructor's explicit permission, after you have explained your reasons (Part 0).
- **New:** a misleading plot detection test between partners, and a model case that decides regression or classification, and linear or forest.

One person teams: *If you are a team of one, this notebook counts as both your individual and your group work. Complete everything the individual handout asks for inside this notebook (label your figures A1 to A4 and MA), then complete this assignment.* ***Submit the same files to both the Phase 3 Individual and the Phase 3 Group assignments on Canvas.*** *It is graded once against each rubric. Wherever this handout says "both members", read it as "you". The specific adjustments are marked* ***Solo:*** *in each part.*

Part 0: Proposal Revisit (10 points)

As in Phase 2, this part goes at the top of your notebook. In a markdown cell titled **Part 0: What Changed from Phase 2**, answer all four questions.

1. **Was the research question modified?** Yes or no. If yes, write the Phase 2 version and the Phase 3 version one after the other so the difference is visible.
2. **Was the dataset modified?** Yes or no. There are two kinds of change:
    - *Modifying your current data:* further cleaning or derived columns applied in code on top of `phase2_working.csv`. Examples: removing a sentinel value your histograms exposed, adding a log transformed column, binning a variable, or narrowing the scope to the rows your question is actually about. No permission is needed.
    - *Switching to a different dataset:* allowed only with the instructor's explicit permission, given after you have explained why your current dataset cannot support your question. State when permission was given, summarize what you explained, and show the new dataset's cleaning steps in code along with a short check of the five data properties, since your Phase 2 audit no longer covers it.
3. **How?** If the dataset changed, show the code, print the shape before and after, and save the result as `phase3_working.csv`. Every later part of this notebook and all of Phase 4 must use that file. If nothing changed, keep using `phase2_working.csv` and say so.
4. **Why?** Three to five sentences tied to evidence. Strong reasons come from what your plots revealed (a spike at 0 or 999, a scope gap, a skew that needs a log), from the five data properties (structure, granularity, scope, temporality, faithfulness), or from Phase 2 feedback. If you changed nothing, do not write "it was fine". Say what you checked in your plots that convinced you the question and data still hold.

Dataset rule: *If you believe your dataset cannot support your question, explain your reasons to the instructor and get explicit permission before you switch. A switch made without that permission will not be accepted, and the notebook will be graded against* `phase2_working.csv`*.*

Part 1: Composite Portfolio (20 points)

1A. Combine (10 points)

Bring both members' Part 1 plots (A1 to A4 and B1 to B4) into this notebook, together with their question, interpretation, and principle check cells. Copying the cells in is fine; this is a composite, not a redo. Then build a coverage table in a markdown cell:

Figure	Member	Slot	Variables	Question it answers	Principle checked
A1	…	1	…	…	…
B3	…	3	…	…	…

Below the table, write three to five sentences on what the eight plots say **together** that no single plot says alone. Where two plots disagree or seem to disagree, say which you trust and why.

1B. Upgrade the Two Weakest (10 points)

As a team, pick the two weakest of the eight plots. For each:

1. Say in one sentence why it is weak, naming a principle (for example, "B2 fails scale: the long right tail squeezes 95 percent of the data into two bars").
2. Rebuild it. Keep the original above the new one so the before and after are visible.
3. Say in one sentence what the upgrade reveals that the original hid.

Useful tools from lecture: a log transform, changing the KDE bandwidth, `alpha` and jitter, conditioning on a third variable with `hue`, switching from a pie or stacked bar to side by side bars, and swapping a rainbow colormap for `viridis`.

Solo: combine your four plots, build the same table, and upgrade the two weakest of the four.

Part 2: Lie Detector (20 points)

Each member built a misleading plot in the individual assignment, using different principles. Now you test whether they work, and whether you can catch them.

2A. The Swap (8 points)

Before reading your partner's explanation (their 2C), look only at their misleading plot and false headline. In a markdown cell titled with your name, write:

1. Whether you believed the headline at first glance, honestly.
2. Which principle you think was broken, and the specific design choice you think does the damage.
3. What you would change to fix it.

Then read their 2C and write one sentence on whether your detection was right, partly right, or wrong. You are graded on the honesty and specificity of the attempt, not on whether you guessed correctly.

Solo: run a cold read instead. Show your misleading plot and headline (only those) to one person outside this course. Record what they concluded and whether they believed the headline, with no names or identifying details. Then write the detection answers above for your own plot as if you were seeing it fresh. As with the Homework 2 surveys, record what really happened. Do not invent a response.

2B. Both Plots, Fixed (6 points)

Bring in both misleading plots (MA and MB), each with its required **MISLEADING ON PURPOSE** label, false headline, honest rebuild, and 2C explanation. The two must use different principles.

Solo: bring in your one misleading plot and its fix. Then, in words only (no plot needed), describe how you would break a second, different principle on the same data and what the false headline would be.

2C. The Checklist (6 points)

Write a short checklist your team will run on every plot in Phase 4 and in your final report. It must include at least one check for each of the four principles (scale, conditioning, perception, context), and each check must be something a reader can actually verify by looking, such as "Does every bar chart start its y axis at zero?" End with two or three sentences on which principle was hardest to catch in the swap, and why you think that is.

The one rule still applies: *Every value in a misleading plot must come from your real working dataset, unaltered. The lie comes from design only. Fabricated or edited values receive zero for Part 2.*

Part 3: Hero Figure (10 points)

Make the single plot that best answers your research question. This is the figure you would put on the first slide of your final presentation. It may be new, or an improved version of one of your eight.

1. It must pass your Part 2C checklist. Show this with a short table: each of the four principles, and one sentence on how the figure meets it.
2. It must explain itself to a stranger. Write a caption of two to four sentences under it that states the finding, the units, the data source, and the number of rows the plot is based on.
3. It may go beyond the four menu slots. Conditioning on a third variable with `hue` or panels, a transformation, or a KDE with a deliberately chosen bandwidth are all welcome, as long as you say why.

Part 4: The Model Case (25 points)

In Phase 4 you will build a model. Here you decide, with evidence, what kind. Each member did a target signal check individually, so you now have two independent reads of the same data. Use the guide below, which is the same one from the individual handout.

The first decision (regression or classification) comes from the **type of your target**. The second decision (a linear model or a forest) comes from the **shape of the relationships** you see in your plots. The two decisions are independent: there are linear classifiers (logistic regression) and forest regressors.

Step 1. Name your target. This is the one column you would predict if you could only predict one. It is usually sitting inside your research question already. If your question is "What drives player wages?", the target is wage.

Step 2. Classify the target.

- Quantitative, where arithmetic means something (price, minutes, score, temperature): this is **regression**. The default starting point is linear regression.
- Qualitative, nominal or ordinal (yes or no, genre, risk tier): this is **classification**. The default starting point is logistic regression for two classes.
- Numbers that are really labels (ZIP codes, jersey numbers, survey codes 1 to 5 with no equal spacing): treat them as qualitative. `int64` does not mean quantitative.

Step 3. Read the shape in your plots.

Signals for a linear model	Signals for a tree based model (random forest)
The scatter of target against a feature looks like a band around a straight line	Steps, thresholds, or plateaus (nothing happens until a feature crosses a value, then everything changes)
A log transform straightens a curved relationship	U shapes or humps that no transformation straightens
When you condition on a category, the lines are roughly parallel	When you condition on a category, the lines cross or flip direction (an interaction)
For classification: the overlaid KDEs or violins of the classes separate cleanly along one feature	For classification: the classes only separate when you combine features, or overlap on every single one
You need to explain each feature's effect in one sentence	Many qualitative features with many levels, and prediction matters more than a one line explanation

Step 4. Check for leakage. A feature leaks if it is computed from the target or recorded after the outcome happened. Common forms: a total that is the sum of parts you also include, a rate built from the counts you are predicting, or a column filled in after the event. A leaking feature makes any model look brilliant and is useless in practice. Drop it now.

Step 5. Name a baseline. For regression, predict the mean of the target for every row. For classification, predict the most common class for every row. Whatever you build in Phase 4 has to beat this.

What is a forest?: *A **linear model** gives every feature one dial: turn up square footage by one unit and the predicted price moves by the same amount every time. A **decision tree** is a flowchart of yes or no questions (Is age over 40? Is region West?) that ends in a prediction. A **random forest** grows hundreds of slightly different trees on random resamples of your rows and lets them vote. It picks up thresholds and interactions without you designing them, but you give up the one sentence explanation a linear model hands you for free. We cover both properly in lecture. For now you only need to recognize which shape your data has.*

4A. Target and Task (5 points)

Name the target column and classify it (quantitative or qualitative, and the finer type). State whether the task is regression or classification. If the two members named different targets or task types in their individual notebooks, show both, then state which one the team is going with and why. If the target needs a small change to fit the question (for example turning a 1 to 5 rating into "4 or above" versus "below 4"), justify it here and make the change in Part 0.

4B. Shape Evidence (8 points)

Show at least **three** plots of the target against features, at least one of which conditions on a qualitative variable. You may reuse plots from Part 1, the individual target checks, or Part 3; reference them by figure label instead of making them twice. For each, write one or two sentences naming which signal from the Step 3 table you see.

At least one plot must test a transformation: plot the target (or a key feature) on a log scale and say whether the relationship becomes straighter. If it does not apply, for example because the target is qualitative, say why.

4C. Linear or Forest (5 points)

Fill in this decision table for your project, then make the call.

Question	What our plots show (cite figures)	Points toward
Is the relationship roughly straight, or can a transformation make it so?	…	Linear or forest
Are there thresholds, steps, or plateaus?	…	…
Do conditioned lines stay parallel or cross?	…	…
Do we need to explain each feature's effect in a sentence?	…	…

4D. Leakage and Baseline (4 points)

List every feature you are dropping because it leaks, with one sentence each on how it is connected to the target. Then compute the baseline with Pandas (the mean of the target for regression, or the share of the most common class for classification) and state it as the number your Phase 4 model must beat.

4E. The Case (3 points)

One paragraph, written as an argument:

- **Claim:** "Our target is …, so this is a … task, and we will start with a … model."
- **Evidence:** the two strongest figures, by label.
- **Counterargument:** the best reason someone might choose the other model family.
- **What would change our minds:** what result in Phase 4 would make you switch.

Part 5: Honest Data Story (10 points)

Write 250 to 400 words, in a markdown cell, that tells the true story of your data so far to someone who has never seen your project. Use at least three of your figures, referenced by label, including the hero figure. The story must include one thing your data **cannot** tell you, and why (a scope gap, a missing variable, a sampling issue, or a correlation you cannot call causal). This is the honest counterpart to the false headlines from Part 2: same data, no tricks.

Part 6: Reproducibility, README, and Contributions (5 points)

- The notebook runs from top to bottom after Kernel, Restart and Run All, using only the files you submit. If you submitted a snapshot, the notebook runs on the snapshot and the README says so.
- A README (markdown cell at the top or a separate `README.md`) listing the files, the working dataset used, and the order to run things in.
- A contributions statement: two or three sentences per member on what they did in the group notebook, beyond their individual assignment.
- An AI use statement naming any tool used and what for. The Phase 2 AI Ethics Clearance still applies to any data you paste into an AI tool. Parts 2, 4E, and 5 must be in your own words.

What to Submit

Submit to the **Phase 3 Group** assignment on Canvas (one submission per team):

- The group notebook, named `GroupNumber_FirstName1_FirstName2_Phase3.ipynb`
- An HTML export of the same notebook
- The working dataset you used (`phase2_working.csv`, or `phase3_working.csv` if you changed it in Part 0), or your snapshot CSV if the dataset is too large to submit (see below)
- The README, if it is a separate file

Solo: name the notebook `GroupNumber_FirstName_Phase3.ipynb` and submit the same files to **both** the Phase 3 Individual and the Phase 3 Group assignments.

If Your Dataset Is Too Large to Submit

Some working datasets are too large to upload. If yours is one of them, do not submit the full file. Instead, capture a snapshot of it, save the snapshot as a CSV, and have your notebook load and parse that CSV. Your whole notebook must then run from top to bottom on the snapshot alone.

Step 1: Capture the snapshot (once, in a separate cell or script, then comment it out). Draw rows at random with a fixed seed so the snapshot is reproducible:

```
full = pd.read_csv("phase2_working.csv")      # or phase3_working.csv
snapshot = full.sample(n=20000, random_state=577)
snapshot.to_csv("phase3_snapshot.csv", index=False)
```

If your target is qualitative and one class is rare, sample within each class instead, so the rare class does not vanish:

```
snapshot = full.groupby("target_col").sample(frac=0.10, random_state=577)
```

Do not use `head()`. The first rows of a file are almost never representative, since files are often sorted by date, ID, or category.

Step 2: Load and parse the snapshot. Read the snapshot back in and check that every column came back as the right type. Saving to CSV throws away dtypes, so dates return as strings and codes like ZIP lose their leading zeros unless you parse them again:

```
df = pd.read_csv("phase3_snapshot.csv",
                 parse_dates=["date_col"],
                 dtype={"zip_col": str})
df.shape
df.dtypes
```

Step 3: Document it. In a markdown cell right after loading, state:

- The shape and file size of the full working dataset, and the shape of the snapshot
- How the snapshot was drawn (random or by class), the sample size, and the seed
- One or two sentences on what the snapshot could miss, such as very rare categories or extreme values

The snapshot must still have at least 1,000 rows, and your team should use the same snapshot file in every notebook, individual and group. Submit the snapshot CSV in place of the full file.

Grading Criteria

Criterion	Points
Part 0: Question and dataset change status stated, changes shown in code with shapes before and after, working file named correctly, instructor permission noted for any dataset switch, and an evidence based reason (“it was fine” earns no credit for the reason)	10
Part 1A: All individual plots combined with their cells, complete coverage table, and a synthesis of what the plots show together	10
Part 1B: Two weakest plots identified with a named principle, rebuilt with before and after visible, and the gain explained	10
Part 2A: Honest, specific detection attempt from each member (or a real cold read, for solo teams), followed by the check against 2C	8
Part 2B: Both misleading plots present with labels, headlines, honest fixes, and explanations, using two different principles (solo: one plot plus a written second principle)	6
Part 2C: Checklist with at least one visually verifiable check per principle, and the reflection on which was hardest to catch	6
Part 3: Hero figure that answers the research question, a principle table, and a caption a stranger could follow (finding, units, source, row count)	10
Part 4A: Target and task type correct, and any disagreement between members resolved with a reason	5
Part 4B: At least three target versus feature plots, one conditioned, one testing a transformation, each tied to a named signal	8
Part 4C: Completed decision table citing figures, and a clear linear or forest call	5
Part 4D: Leakage features identified and dropped with reasons, and a baseline computed in Pandas	4
Part 4E: Case paragraph with claim, evidence, counterargument, and what would change your minds	3
Part 5: Honest data story of 250 to 400 words using three or more figures, including one thing the data cannot tell you	10
Part 6: Runs from top to bottom (on the documented snapshot, if used), README, contributions statement, and AI use statement	5
Total	**100**

Strong visual evidence here makes Phase 4 far easier. If your plots already show you that the relationship bends at a threshold, you will not spend a week wondering why your straight line keeps missing.